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Predictive Modeling of Generative AI Inference Cost Across Hardware Architectures

Yongjun Kim

***Abstract*—Deploying generative AI models across heterogeneous hardware requires accurate cost estimation at design time, yet inference energy and latency vary dramatically with model architecture, output configuration, and platform characteristics. This brief presents a systematic framework for predicting generative AI inference cost from architectural and hardware parameters without executing the target workload. Using 4,133 energy measurements spanning 7 generative models (autoregressive and diffusion-based) across three hardware platforms (Apple M4 Pro, NVIDIA A100, NVIDIA H100), individual measurements from 137 unique configurations ($\sim$30.2 repetitions each) are used to train and evaluate five regression methods under three cross-validation strategies: standard k-fold, leave-one-hardware-out (LOHO), and leave-one-model-out (LOMO). SVR with log-transformed features achieves $R^2 = 0.923$ and MAPE of 14.2% under k-fold CV, demonstrating that accurate cost prediction is feasible when the target hardware is known. However, LOHO-CV reveals a fundamental generalization gap: predicting energy on entirely unseen hardware degrades to $R^2 < 0.1$, exposing the limits of black-box models for cross-platform extrapolation. SHAP analysis uncovers architecture-dependent cost drivers: hardware thermal design power dominates for autoregressive models (mean $|\text{SHAP}| = 0.83$), while output complexity and diffusion steps dominate for diffusion models. These findings provide circuit and system designers with actionable guidance for hardware-aware inference cost estimation and workload-to-platform mapping.**

***Index Terms*—Energy modeling, generative AI, inference cost prediction, hardware-aware design, cross-platform estimation, SHAP analysis.**

I. Introduction

The proliferation of generative AI—spanning large language models (LLMs), image diffusion models, video generators, and audio synthesizers—has created an urgent need for hardware-aware inference cost modeling. Data center operators, edge device designers, and system architects must estimate energy consumption and latency *before* committing to specific hardware, yet the design space of models, configurations, and platforms is vast [1], [2].

Two fundamental architectural paradigms dominate generative AI: *autoregressive* (AR) models that generate tokens sequentially, and *diffusion* models that iteratively denoise through multiple steps. These paradigms exhibit fundamentally different computational profiles—AR cost scales with output token count while diffusion cost scales with the product of spatial resolution and step count—making unified cost prediction challenging [3].

System-level energy modeling is a well-established topic in circuits and systems research, spanning processor power estimation [4], DRAM energy models [5], accelerator design space exploration [6], and neural network accelerator power prediction [7]. However, existing cost models for neural network inference focus primarily on convolutional or transformer architectures in classification tasks [8], [9], with limited coverage of generative workloads that exhibit order-of-magnitude variation in output complexity. Recent LLM energy profiling efforts [10]–[12] are restricted to single modalities and platforms.

This brief addresses the question: *Can generative AI inference cost be predicted from model and hardware parameters alone, without running the workload on the target platform?* The contributions of this work are:

1) A unified dataset of 4,133 energy and latency measurements across seven generative models, four modalities (text, image, video, music) with varying batch configurations, and three hardware platforms spanning a 17.5$\times$ range in TDP.
2) A systematic comparison of five regression methods under three cross-validation strategies, revealing the generalization gap between within-platform and cross-platform prediction.
3) SHAP-based analysis showing that the dominant cost drivers differ fundamentally between AR and diffusion architectures, providing actionable insights for system designers.
4) Practical guidelines for hardware-aware cost estimation at design time, with implications for SoC power budgeting and hardware design space exploration.

II. Background and Related Work

A. System-Level Energy Modeling

Architectural power estimation has a long history in the circuits community. Wattch [4] pioneered cycle-accurate power modeling for microprocessors, enabling design-time power exploration. Later work extended this to memory subsystems [5] and heterogeneous SoCs [13]. For neural network accelerators, Eyeriss [6] introduced dataflow-level energy analysis, and subsequent work developed layer-wise energy models based on operation counting [8], [9]. The roofline model [14] provides hardware-centric performance bounds based on arithmetic intensity, offering a complementary analytical approach to the empirical modeling pursued in this work. Machine learning approaches to hardware cost estimation [7], [15] have shown promise for design space exploration, but focus on training rather than inference workloads.

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The cost model proposed here operates at the *system level*, predicting total inference energy from high-level specifications, complementing microarchitectural tools such as Wattch [4] that provide detailed bottom-up power analysis. System-level models enable rapid design-time estimates for early-stage hardware selection and SoC power budgeting.

B. Generative AI Inference Cost

AR models (GPT, Phi, Mistral, MusicGen) generate tokens sequentially with cost scaling linearly with output length [16], [17]. Diffusion models (Stable Diffusion, SDXL, AnimateDiff) perform iterative denoising with per-step cost proportional to model size and spatial resolution [18], [19]. Recent studies have characterized LLM inference energy [10], carbon cost [11], [12], and edge efficiency [20], but these *report* measured costs rather than *predicting* them from specifications. This work fills that gap by building predictive models that attempt to generalize across hardware platforms.

C. Explainable ML for Hardware

SHAP [21] provides theoretically grounded feature attribution and has been applied to identify power consumption drivers in hardware [22]. Here, SHAP decomposes inference cost into interpretable architecture and hardware components.

III. Methodology

A. Measurement Infrastructure

Inference energy measurements were collected using platform-specific instrumentation: NVIDIA GPUs via `nvidia-smi` power sampling at $100\,\mathrm{ms}$ intervals, and Apple M4 Pro via `powermetrics` at $500\,\mathrm{ms}$ resolution. Each configuration was repeated approximately 30 times to capture run-to-run variance. The 4,133 individual measurements are aggregated to 137 configuration-level means (averaging $\sim$30.2 repetitions per configuration) prior to modeling, ensuring that repetitions of the same configuration do not appear in both training and test folds during cross-validation. The energy measurements in this study were collected as part of a broader measurement campaign also reported in a companion study examining energy crossover effects; the present work addresses the distinct research question of predictive cost modeling.

B. Models, Hardware, and Configurations

Table I summarizes the seven generative models and three hardware platforms. The dataset spans four modalities (text, image, video, music) with varying batch configurations: token counts from 32 to 512, image resolutions from 128×128 to 640×640, video frame counts from 4 to 16, diffusion steps from 5 to 50, and batch sizes from 1 to 16. The dataset comprises 4,133 individual measurements across 137 unique configurations ($\sim$30.2 repetitions per configuration), with energy ranging from $3.1\,\mathrm{J}$ to $1728\,\mathrm{J}$ and latency from $0.16\,\mathrm{s}$ to $71\,\mathrm{s}$.

TABLE I
Models, Hardware Platforms, and Coverage

Model	Type	Params	Mod.	M4P	A100	H100
Phi-3-mini-4k	AR	3.8 B	Text	✓	✓	✓
Mistral-7B	AR	7.2 B	Text	–	✓	–
MusicGen-small	AR	0.6 B	Music	✓	✓	–
MusicGen-medium	AR	1.5 B	Music	✓	✓	–
SD-v1-5	Diff	0.9 B	Image	✓	✓	✓
SDXL-base-1.0	Diff	3.5 B	Image	✓	✓	✓
AnimateDiff-v1.5	Diff	0.9 B	Video	✓	✓	✓
Hardware	**TDP**	**Mem. BW**	**Compute**			
Apple M4 Pro	40 W	273 GB/s	20 GPU cores			
NVIDIA A100 40GB	400 W	1555 GB/s	6912 CUDA cores			
NVIDIA H100 80GB	700 W	3350 GB/s	14592 CUDA cores			

Checkmarks indicate measured combinations. Not all models were tested on all platforms; Mistral-7B was measured only on A100, and MusicGen variants only on M4 Pro and A100. This affects LOHO and LOMO evaluation as discussed in the text.

C. Feature Engineering

Eight input features are engineered from model specifications and hardware datasheets, requiring *no runtime profiling*:

- **Model:** parameter count (billions), architecture type (AR $=0$, Diffusion $=1$)
- **Hardware:** thermal design power (TDP, watts), compute units (GPU cores or CUDA cores), memory bandwidth (GB/s)
- **Configuration:** output complexity (tokens for text/music, pixels for images, pixel$\times$frames for video), diffusion steps (1 for AR), batch size

Output complexity provides a unified scalar that captures the workload magnitude across heterogeneous modalities. All continuous features are log-transformed via $\log(1+x)$ to handle the multi-scale ranges (TDP: 40–700 W; compute: 20–14592 units; energy: 3–1728 J). Target energy is also log-transformed for training; predictions are inverse-transformed before evaluation.

D. Regression Methods

Five regression approaches spanning parametric, kernel, and ensemble methods are evaluated: Linear Regression, Ridge (α=1), Random Forest (200 trees, max depth 12), XGBoost (200 boosters, max depth 8, learning rate 0.1), and SVR (RBF kernel, C=100). For SVR, features are standardized; tree-based models operate on log-transformed features directly. Hyperparameters were set to commonly used defaults from the scikit-learn and XGBoost documentation; no hyperparameter tuning was performed on the evaluation data.

E. Cross-Validation Strategies

Three CV strategies of increasing difficulty are employed:

1) **5-Fold CV:** random splits across all configurations; tests interpolation within the training distribution.
2) **Leave-One-Hardware-Out (LOHO):** train on two platforms, predict the third; tests cross-platform extrapolation—the key experiment for circuits/systems applications.

TABLE II
5-FOLD CROSS-VALIDATION RESULTS (ENERGY)

Method	R^2	RMSE (J)	MAPE (%)
Linear	0.571	243.8	45.1
Ridge	0.374	294.5	67.7
RF	0.761	182.1	36.8
XGBoost	0.839	149.4	34.1
SVR	**0.923**	103.3	14.2

TABLE III
LEAVE-ONE-HARDWARE-OUT CV RESULTS (ENERGY)

Method	R^2	RMSE (J)	MAPE (%)
Linear	-0.59	469.2	333.8
Ridge	-0.80	499.5	397.4
RF	0.05	362.2	208.7
XGBoost	-0.02	375.2	143.9
SVR	-0.28	421.0	392.0

3) **Leave-One-Model-Out (LOMO):** train on six models, predict the seventh; tests generalization to unseen architectures.

LOHO simulates the realistic scenario where a designer wants to estimate inference cost on hardware that has not yet been profiled. This is the most demanding test and the primary experiment of this work.

IV. RESULTS AND ANALYSIS

A. Within-Platform Prediction (5-Fold CV)

Table II reports 5-fold CV results for energy prediction using log-transformed features and log-space training (with evaluation in raw energy space). SVR achieves $R^2 = 0.923$ with MAPE of 14.2%, demonstrating that the eight engineered features capture the essential cost structure when properly transformed. Linear models perform moderately ($R^2 \approx 0.57$), while tree-based ensembles achieve $R^2 \approx 0.76$–0.84. The log transformation is critical: raw-feature models (not shown) achieve substantially lower accuracy, confirming the multiplicative nature of the inference cost function. When the target hardware is represented in the training data, accurate cost prediction is feasible with simple features and standard regression methods. Because individual repetitions are aggregated to configuration-level means prior to cross-validation, each fold contains distinct configurations with no data leakage from repeated measurements; the LOHO and LOMO strategies, which group by hardware platform or model respectively, provide more conservative generalization estimates.

B. Cross-Platform Prediction (LOHO-CV)

Table III presents the critical LOHO-CV results. All models suffer dramatic performance degradation when predicting on unseen hardware. Even the best performer (RF) achieves only $R^2 = 0.05$ overall. This exposes a fundamental *generalization gap*: models that achieve over 90% accuracy within known platforms fail when extrapolating to new hardware.

The per-platform breakdown (Table IV) reveals pronounced asymmetry. The A100 is most predictable when held out (best MAPE of 33.5%), as its specifications fall between the M4 Pro and H100, enabling interpolation. The M4 Pro is hardest to predict (MAPE $> 375\%$ for all models), likely because its unified memory architecture and low TDP (40 W vs. 400 W–700 W for GPUs) create a qualitatively different energy profile from discrete GPUs—models trained on high-TDP platforms systematically overestimate M4 Pro consumption.

TABLE IV
PER-PLATFORM LOHO ENERGY MAPE (%)

Method	M4 Pro	A100	H100
Linear	919.0	33.5	263.2
Ridge	993.1	33.5	352.8
RF	425.6	40.0	209.3
XGBoost	375.8	58.4	100.3
SVR	1077.0	40.1	309.5

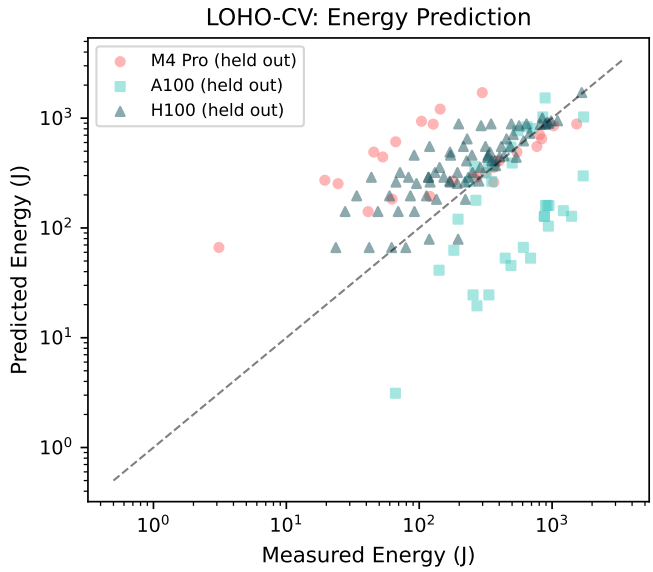

Fig. 1. LOHO-CV predicted vs. measured inference energy using XGBoost in log-feature space. Each hardware platform is predicted by a model trained exclusively on the other two. A100 predictions (interpolation) closely follow the diagonal; M4 Pro predictions (extrapolation) show systematic overestimation.

Fig. 1 visualizes the LOHO predictions using XGBoost trained in log-transformed feature space. Predictions cluster near the diagonal for A100 (interpolation regime) but show systematic bias for M4 Pro (extrapolation regime), confirming that the gap is structural rather than stochastic.

C. Cross-Model Prediction (LOMO-CV)

LOMO-CV evaluates generalization to unseen model architectures. SVR achieves the best overall $R^2 = 0.28$, outperforming tree-based methods. Performance varies by held-out model: AnimateDiff is well-predicted (MAPE $\approx$ 28–30%) because other diffusion models provide similar training signal, while MusicGen-small proves challenging despite MusicGen-medium (same family) being in training—the parameter size gap (0.6 B vs. 1.5 B) limits generalization.

D. Feature Importance via SHAP Analysis

Fig. 2 presents SHAP feature importance decomposed by architecture type. Three key findings emerge:

(1) Overall: Output complexity, diffusion steps, and model size are the top-three cost drivers, followed by hardware TDP. Batch size has minimal impact at the scales tested.

(2) AR models: Hardware TDP dominates (mean $|\text{SHAP}|=0.83$), followed by output complexity (0.54). AR inference is *hardware-bound*: the platform's power envelope determines total energy more than model architecture. This aligns with the observation that AR models maintain near-constant power draw during sequential token generation [10]. This implies that power management strategies (DVFS, power gating) can yield significant energy savings for AR workloads.

(3) Diffusion models: Output complexity (0.57) and diffusion steps (0.43) dominate, with hardware TDP being secondary (0.11). Diffusion inference is *workload-bound*: the iterative denoising loop saturates hardware, making workload parameters the dominant cost factors.

This dichotomy has direct implications for system designers: optimizing AR workloads benefits most from hardware power management (lower-TDP platforms save energy proportionally), while diffusion workloads benefit from algorithmic improvements such as step reduction and distillation [23].

E. Analytical Cost Formula

A log-linear analytical model of the form is also fitted:

$$\log(1+E) = \beta_0 + \sum_{i=1}^{7} \beta_i \cdot f_i + \beta_8 \cdot \mathbb{1}_{\text{Diff}} \tag{1}$$

where f_i are log-transformed features and $\mathbb{1}_{\text{Diff}}$ is the architecture indicator. The fitted coefficients reveal the multiplicative cost structure: energy scales as $E \propto P^{0.46} \cdot C^{0.64} \cdot S^{1.06} \cdot \text{TDP}^{-0.32} \cdot \text{BW}^{-0.95}$, where P is model size, C is output complexity, S is diffusion steps, and the negative TDP/bandwidth exponents reflect that higher-capability hardware completes work faster, reducing energy duration. This formula achieves $R^2=0.61$ in-sample and MAPE of 489% under LOHO-CV, confirming that the cross-platform extrapolation challenge persists even with explicitly structured models. The analytical form is analogous to the roofline model [14] in spirit—both capture hardware–workload interactions through simple parametric relationships—though the roofline operates on arithmetic intensity while the cost formula operates on high-level specifications.

F. Discussion: The Generalization Gap and Systems Implications

The contrast between 5-fold CV ($R^2=0.923$) and LOHO-CV ($R^2<0.1$) constitutes the central finding of this work. This gap arises from three factors: (1) with only three hardware platforms, LOHO-CV evaluates three specific train/test splits rather than a statistically powered sample—the result demonstrates failure on *these* platform pairs but cannot claim generality beyond them; (2) the three platforms differ qualitatively (unified vs. discrete memory architecture, 17.5× TDP range, different memory hierarchies), not just quantitatively; (3) tree-based models fundamentally cannot extrapolate beyond the range of training data, and with hardware features at the extremes of the training distribution, LOHO prediction becomes extrapolation rather than interpolation. Additionally, incomplete model coverage across platforms (Table I) means that LOHO simultaneously tests cross-platform *and* cross-model transfer in some cases, conflating two sources of difficulty.

From a circuits and systems perspective, these results have several practical implications. First, *hardware-specific calibration is essential*: even ∼10 profiling runs on the target platform would anchor the model substantially, motivating a "few-shot hardware adaptation" workflow.

Second, the SHAP-identified cost driver dichotomy enables *targeted optimization*: AR workloads benefit from power-efficient hardware selection (since TDP dominates), while diffusion workloads benefit from algorithmic step reduction [23], [24]. For SoC power budgeting, AR inference power can be estimated primarily from TDP, while diffusion inference requires workload characterization. Integrating these findings with microarchitectural tools (Wattch [4], Accelergy [7]) could enable hierarchical power estimation: rapid top-down system-level estimates with detailed bottom-up validation.

Third, the asymmetric LOHO performance—where interpolation (A100 held out) succeeds but extrapolation (M4 Pro held out) fails—suggests that *hardware architecture class* matters more than individual specifications. Discrete GPU platforms with CUDA cores and HBM form one class; unified-memory SoCs with integrated GPU cores form another. Future cost models should incorporate explicit hardware class indicators or employ domain adaptation techniques to bridge these architectural gaps.

G. Practical Design Guidelines

Based on the analysis, the following guidelines are offered for system designers estimating generative AI inference cost:

1) **Same platform:** If even partial profiling data exists on the target hardware, SVR with log-features provides accurate prediction ($R^2>0.9$, MAPE $<15\%$) from only eight easily obtainable features.
2) **Similar platform:** For hardware within the same architectural class (e.g., A100 → H100), models trained on the source can provide reasonable estimates (MAPE ∼35–60%).
3) **Different platform class:** Cross-class prediction (e.g., GPU → SoC) requires hardware-specific calibration. The analytical formula (1) provides order-of-magnitude estimates; accurate prediction requires on-target measurement.
4) **New model architecture:** The LOMO results suggest that prediction accuracy depends on the presence of architecturally similar models in the training set. AR-to-AR and diffusion-to-diffusion transfer is more reliable than cross-paradigm prediction.

V. Conclusion

A predictive cost modeling framework for generative AI inference across heterogeneous hardware has been presented.

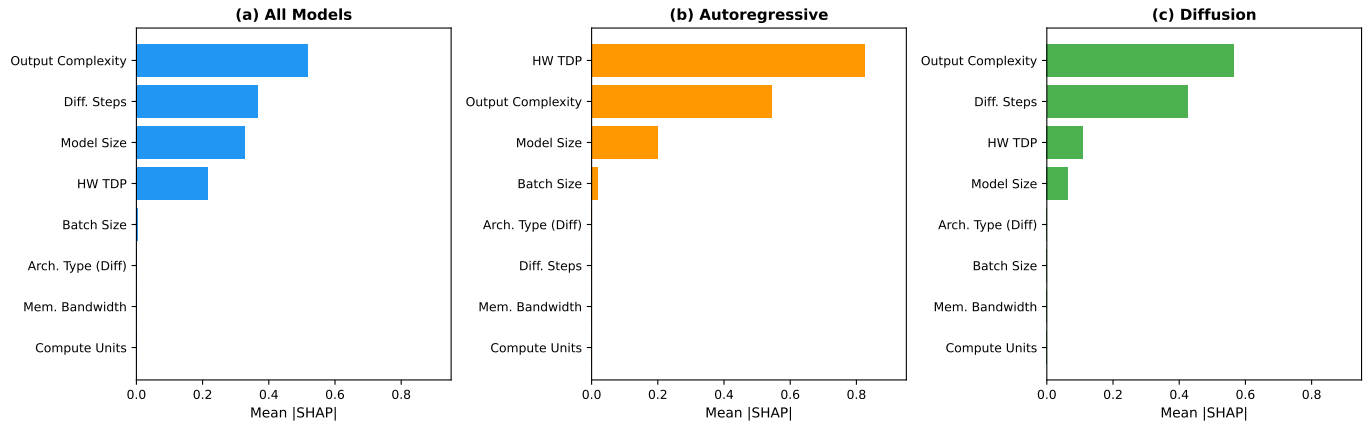

Fig. 2. SHAP feature importance for energy prediction decomposed by architecture type. (a) All models combined: output complexity, diffusion steps, and model size are the top drivers. (b) Autoregressive models only: hardware TDP dominates, revealing hardware-bound behavior. (c) Diffusion models only: output complexity and diffusion steps dominate, revealing workload-bound behavior. All x-axes share the same scale for cross-panel comparison. Features are log-transformed.

The key findings are threefold: (1) within-platform prediction is accurate ($R^2 = 0.923$, MAPE $\approx 14.2\%$), validating that feature-based cost modeling captures the essential structure of inference energy consumption; (2) cross-platform prediction via LOHO-CV exposes a fundamental generalization gap ($R^2 < 0.1$), revealing that training on a small number of hardware platforms is insufficient for reliable extrapolation—a critical insight for the circuits community seeking universal cost models; (3) SHAP analysis uncovers a hardware-bound vs. workload-bound dichotomy between AR and diffusion architectures, providing architecture-specific optimization guidance.

For circuit and system designers, these results suggest combining a general-purpose model with lightweight hardware-specific calibration. The cost drivers provide actionable guidance: power management for AR workloads, step reduction for diffusion workloads.

Limitations and future work: The three-platform scope limits its LOHO generalization assessment: with $n=3$ leave-one-out splits, the LOHO results characterize specific platform pairs rather than establishing a general law. The 137 configuration-level data points, while sufficient for the eight-feature models evaluated, would not support higher-dimensional approaches. Incomplete model coverage across platforms (Table I) introduces asymmetry in LOHO evaluation. Additionally, log-space training optimizes relative rather than absolute error; high-energy configurations may have larger absolute residuals than the MAPE suggests. Future work should expand to additional platforms (FPGAs, NPUs, quantized variants) to enable statistically meaningful cross-platform analysis; apply transfer/meta-learning for few-shot hardware adaptation; and integrate with co-design tools [9] for energy-aware design space exploration.

Data and code availability: The measurement data and analysis code are available from the author upon reasonable request.

AI use declaration: AI-assisted tools were used for code review, manuscript editing, and LaTeX formatting. All experimental design, data collection, analysis, and scientific interpretation were performed by the author.

References

[1] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "Carbon emissions and large neural network training," *arXiv preprint arXiv:2104.10350*, 2021.

[2] C.-J. Wu, R. Raghavendra, U. Gupta, B. Acun, N. Ardalani, K. Maeng, G. Chang, F. Aga, J. Huang, C. Bai *et al.*, "Sustainable AI: Environmental implications, challenges and opportunities," *Proc. Mach. Learn. Syst.*, vol. 4, pp. 795–813, 2022.

[3] Y. Cao, S. Li, Y. Liu, Z. Yan, Y. Dai, P. S. Yu, and L. Sun, "A comprehensive survey of AI-generated content (AIGC): A history of generative AI from GAN to ChatGPT," *ACM Comput. Surv.*, vol. 56, no. 10, pp. 1–38, 2024.

[4] D. Brooks, V. Tiwari, and M. Martonosi, "Wattch: A framework for architectural-level power analysis and optimizations," in *Proc. Int. Symp. Comput. Archit. (ISCA)*, 2000, pp. 83–94.

[5] K. Chandrasekar, C. Weis, Y. Li, S. Goossens, M. Jung, and N. Wehn, "DRAMPower: Open-source DRAM power and energy estimation tool," *URL: http://www.drampower.info*, 2012.

[6] Y.-H. Chen, T. Krishna, J. S. Emer, and V. Sze, "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 127–138, 2017.

[7] Y. N. Wu, J. S. Emer, and V. Sze, "Accelergy: An architecture-level energy estimation methodology for accelerator designs," in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Design (ICCAD)*, 2019, pp. 1–8.

[8] T.-J. Yang, Y.-H. Chen, and V. Sze, "Designing energy-efficient convolutional neural networks using energy-aware pruning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 5687–5695.

[9] E. Cai, D.-C. Juan, D. Stamoulis, and D. Marculescu, "NeuralPower: Predict and deploy energy-efficient convolutional neural networks," *arXiv preprint arXiv:1710.05420*, 2017.

[10] S. Samsi, D. Zhao, J. McDonald, B. Li, A. Michaleas, M. Jones, W. Bergeron, J. Kepner, D. Tiwari, and V. Gadepally, "From words to watts: Benchmarking the energy costs of large language model inference," *arXiv preprint arXiv:2310.03003*, 2023.

[11] A. A. Chien, L. Lin, H. Nguyen, V. Rao, T. Sharma, and R. Wijayawardana, "Reducing the carbon impact of generative AI inference (today and in 2035)," *arXiv preprint arXiv:2309.02427*, 2023.

[12] A. S. Luccioni, Y. Jernite, and E. Strubell, "Power hungry processing: Watts driving the cost of AI deployment?" *Proc. ACM Conf. Fairness, Account., Transp. (FAccT)*, pp. 85–99, 2024.

[13] S. K. Rethinagiri, O. Palomar, J. A. Moreno, O. Unsal, and A. Cristal, "System-level power estimation tool for embedded processor based platforms," in *Proc. ACM Int. Conf. Compilers, Archit., Synth. Embed. Syst. (CASES)*, 2014, pp. 1–10.

[14] S. Williams, A. Waterman, and D. Patterson, "Roofline: An insightful visual performance model for multicore architectures," *Commun. ACM*, vol. 52, no. 4, pp. 65–76, 2009.

[15] A. Boroumand, S. Ghose, Y. Kim, R. Ausavarungnirun, E. Shiu, M. Thottethodi, T. N. Vijaykumar, and O. Mutlu, "Google workloads for consumer devices: Mitigating data movement bottlenecks," in *Proc. Int. Conf. Archit. Support Program. Lang. Oper. Syst. (ASPLOS)*, 2018, pp. 316–331.

[16] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 1877–1901, 2020.

[17] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei, "Scaling laws for neural language models," *arXiv preprint arXiv:2001.08361*, 2020.

[18] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10 684–10 695.

[19] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 6840–6851.

[20] R. Desislavov, F. Martínez-Plumed, and J. Hernández-Orallo, "Trends in AI inference energy consumption: Beyond the performance-vs-parameter laws of deep learning," *Sustain. Comput.: Inform. Syst.*, vol. 38, p. 100857, 2023.

[21] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4765–4774.

[22] Z. Zhang, X.-D. Zhang, and P. Yang, "Machine learning based power modeling of multicore processors using shap analysis," *IEEE Access*, vol. 8, pp. 185 245–185 256, 2020.

[23] T. Salimans and J. Ho, "Progressive distillation for fast sampling of diffusion models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[24] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, "Consistency models," *arXiv preprint arXiv:2303.01469*, 2023.